CYBER ATTACK TRAFFIC CLASSIFICATION (CIC-IDS2017)

Target Classification Labels & Threat Taxonomy: Detailed operational definitions and security context for all 15 evaluated network traffic categories.

#	Target Class Label	Threat Category	Security Definition & Attack Behavior
1	`BENIGN`	**Normal Traffic**	Legitimate, uncompromised network background traffic including normal web browsing (HTTP/HTTPS), DNS queries, file transfers, and streaming services without malicious intent.
2	`Bot`	**Botnet Command & Control**	Traffic generated by compromised host machines communicating with an external Command & Control (C2) server (simulated using the Ares Botnet). Characterized by periodic beaconing and remote execution polling.
3	`DDoS`	**Distributed Denial of Service**	High-volume, coordinated traffic flood originated from multiple distributed sources aimed at saturating network bandwidth and depleting server processing resources, causing widespread denial of service.
4	`DoS GoldenEye`	**Application Layer DoS**	Layer-7 HTTP denial of service tool that exhausts server worker threads and socket pools by keeping persistent HTTP Keep-Alive connections open with randomized headers and cache-busting requests.
5	`DoS Hulk`	**Application Layer DoS**	High-throughput HTTP flood tool (Hope Overwritten by Lost Kraven) designed to bypass caching engines, direct requests straight to the web server daemon, and exhaust backend memory and connection pools.
6	`DoS Slowhttptest`	**Slow-Rate DoS**	Slow-rate application DoS attack that transmits partial HTTP request headers or message bodies at extremely slow transmission rates, holding server concurrent connection slots occupied until server starvation occurs.
7	`DoS slowloris`	**Slow-Rate DoS**	Low-bandwidth denial of service tool that opens numerous connections to the target web server and sends incomplete HTTP headers at periodic intervals to prevent connection timeouts without generating volumetric traffic.
8	`FTP-Patator`	**Brute Force Authentication**	Multi-threaded, automated credential dictionary attack directed against the File Transfer Protocol (FTP, port 21) server to guess valid user credentials via rapid, systematic login attempts.
9	`Heartbleed`	**Memory Exploitation (CVE-2014-0160)**	Critical OpenSSL TLS heartbeat memory disclosure vulnerability where malicious, oversized heartbeat request packets force the server to leak 64KB chunks of contiguous process memory (including private keys and credentials).
10	`Infiltration`	**Privilege Escalation & Pivot**	Post-exploitation lateral movement where an attacker compromises a victim machine inside the network perimeter (e.g., via malicious PDF/vulnerability), establishes a backdoor, scans the local subnet, and attempts privilege escalation.
11	`PortScan`	**Network Reconnaissance**	Systematic scanning of target TCP/UDP port ranges to identify open listener services, operating system fingerprints, and vulnerable software versions before launching targeted exploits.
12	`SSH-Patator`	**Brute Force Authentication**	High-speed automated brute force dictionary attack targeting the Secure Shell (SSH, port 22) service attempting multiple password combinations to gain unauthorized remote root or shell access.
13	`Web Attack - Brute Force`	**Web Application Attack**	Automated authentication brute forcing against web application login forms, content management systems (CMS), and HTTP basic authentication gateways to guess passwords.
14	`Web Attack - Sql Injection`	**Web Application Attack**	Injection of malicious SQL payload fragments into user-controlled input fields or URL parameters to manipulate database queries, bypass authentication, and exfiltrate database records.
15	`Web Attack - XSS`	**Web Application Attack**	Cross-Site Scripting attack where malicious JavaScript payloads are injected into web application inputs to execute arbitrary clientside scripts inside the browsers of other authenticated users.